

The NASA Observing Program: A High-Resolution Microwave Survey

Members: Michael Klein (coordinator), John Billingham, David Milne, Jill Tarter

Introduction

The purpose of the NASA High Resolution Microwave Survey (HRMS) is to search for evidence of extraterrestrial technology, most notably radio transmissions emanating from extrasolar planets. If such were detected, it would be reasonable to assume that we had also detected evidence of intelligent extraterrestrial life. The HRMS project is a part of a broader worldwide program of SETI research activities.

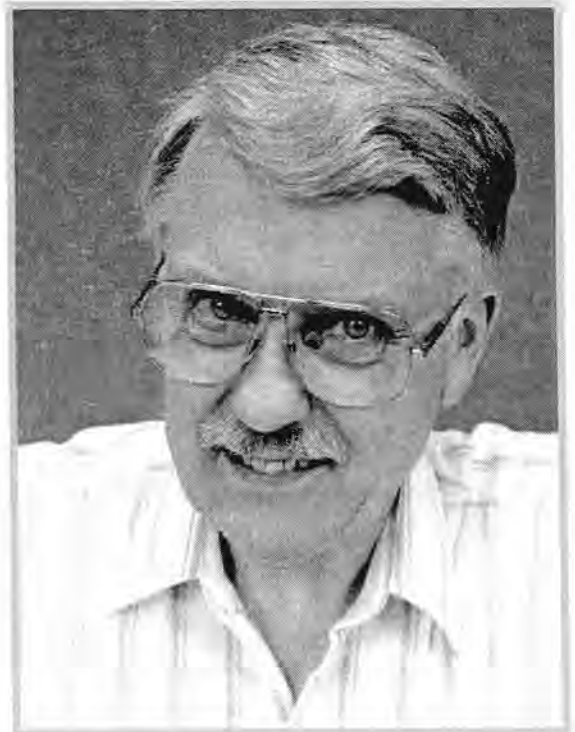
A scientific rationale for the possible existence of extraterrestrial life is now well established, and guides the NASA search. That rationale, with a history of thoughts about extraterrestrial life dating back to antiquity, is traced in Chapter 2. Another section of Chapter 2 deals with the more recent history of SETI in NASA, starting with visionary ideas that first began to influence the Agency in 1969. That vision slowly grew and prompted the emergence of the present comprehensive (yet modest) NASA project to search for radio signals from planetary systems of other stars. That project, now known officially as the High Resolution Microwave Survey (HRMS), was the stimulus for this report.

The search for extraterrestrial intelligence is based upon several assumptions. One is fundamental to all of astronomy; that the same physical processes observable on Earth operate elsewhere in the universe in the same ways

that we see here. Others are applicable to the extraterrestrial civilizations that we seek to detect. These are:

- 1) there is a reasonable likelihood that such civilizations exist;
- 2) any such civilization detected by us is likely to be more technologically advanced than our own.

The first assumption (the probable existence of extraterrestrial civilizations) is discussed in



Mike Klein, of the Jet Propulsion Lab.

Chapter 2 in connection with a formula known as the “Drake equation.” The existence of ETI’s seems to be a reasonable outcome of natural processes operating elsewhere as they have in our solar system. The likely older age of the other civilization is inferred from our knowledge of the speed of transformation of our own society after the advent of industrialization. Only 100 years ago, our own presence in the galaxy was not detectable by radio. It seems unlikely that another civilization just beginning its own SETI research would “tune in” at the moment when our own transmissions began. In reverse, it seems likely that signals coming from elsewhere will be from a civilization that has long since passed the startup of its broadcasting technology. (This expectation is based on a subsidiary assumption, namely that newly-emergent technological civilizations don’t routinely destroy themselves or go off the air for other reasons shortly after acquiring the ability to broadcast.)

The choice of electromagnetic signals as the most promising potential indicators of other civilizations does not stem from assumptions, but is based upon a reasoned rationale that is described in the following.

The Choice – Physical Artifacts, or Electromagnetic Signals?

Any search for evidence of extraterrestrial intelligence (ETI) must begin with questions about ways in which beings of another world might make their presence known to us over the vast distances that separate the stars. A way to begin is to envision manifestations of a distant technology that modifies the natural environment in a way that our instruments can detect. While we are not yet able to detect distant habitable planets, let alone

image them to investigate their surface features or atmospheric chemistries, we may still be able to observe certain consequences of technological activities on distant worlds. Three plausible possibilities are observations that give evidence of (1) transportation, (2) energy production, and (3) communication.

Transportation

A highly advanced technological civilization might be able to send physical objects to our solar system, such as streams of particles, information capsules, robotic spacecraft, or even themselves. A number of interesting conjectures have been offered regarding movements of vehicles across interstellar space (Bracewell, 1976). A fundamental obstacle to this mode of communication is the fact that the distances between stars are so fantastically vast as to dwarf all of the space travel that we have accomplished thus far. Our robotic spaceprobes have traveled several billions of miles to reach the outer planets of our solar system, yet they have traversed less than 0.02 % of the distance to the nearest star (which is about four light years from the Sun). The most distant stars at the opposite edge of the Galaxy are about 80,000 light years from Earth. The laws of physics as we currently understand them impose a fundamental constraint on the speed with which these distances can be crossed. No material object – no particle, no rocket, no starship – can go faster than light. Indeed, if one simply attempts to approach the speed of light, relativistic effects cause the energy requirements to escalate drastically and disproportionately. Our present understanding of physics therefore suggests that, to send an object across interstellar distances, the senders must be willing to invest colossal quantities of energy and endure

voyages that require thousands of years of travel time.

Might an old and highly advanced extrasolar civilization have discovered ways of doing it? We don't know. If we thought so, and if we had reason to believe that they had explored the solar system, we might search for physical artifacts left by their explorations. Such a search might involve examination of the Earth's geological strata for a buried "message capsule," or a comparable search of the planets, or efforts to detect a relict orbiting vehicle in our solar system. However, the certainty that interstellar travel is exorbitantly costly in time and energy (if not flatly impossible) has persuaded most SETI researchers that a search for artifacts is not the best option. A few searches have been made in the directions of the gravitational equilibrium points of the

Earth-Moon system looking for glints of sunlight or radar reflections from artifacts but nothing "big or shiny" has been found. For the most part, NASA scientists have focussed their attention on radio signals, which are thought to be more likely to reveal the existence of ETI's.

Energy Production

Certain side effects of large scale energy production by a highly advanced civilization might be observable from Earth. Several plausible (though highly futuristic) scenarios have been suggested. ETI's might dispose of radioactive waste by launching it into their own star, thereby producing observable effects on the star's spectrum. For example, large-scale energy production via fusion could lead

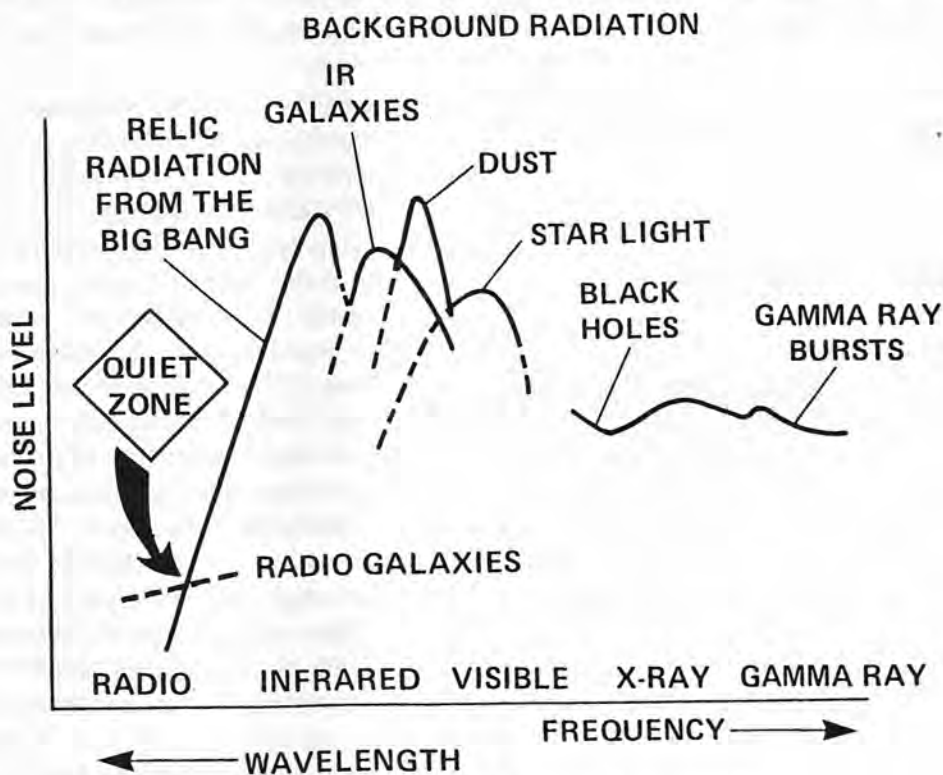


Figure 1. The natural background radiation as a function of wavelength.

to detectable amounts of short-lived (12.5-yr half-life) tritium, a material that is not made by any known natural process. Its presence in the spectrum of light from a star would indicate that some unknown process – perhaps natural, perhaps involving technology – was at work in that neighborhood. In another line of inquiry, some have envisioned that super-civilizations might put an enormous structure in orbit, partially enclosing a central star. Such a structure, known as a Dyson Sphere, could be so large that it captures large fractions of the energy output of the star and re-radiates energy at thermal infrared wavelengths that could be detected by Earth-based telescopes (Dyson 1966).

Sporadic observations conducted over the last 30 years have sought “high tech” artifacts like these. The negative outcome of these searches is hard to interpret, because the scenarios envisioned are far beyond our technical capability. Not knowing what a very advanced civilization can and cannot do, we are unable to ascertain the significance of the negative results.

Interstellar Communication

We cannot seek evidence of the presence of ETI’s who use futuristic communication technologies that are unknown to us. We can, however, search for evidence of any civilizations that use radio signals. Advances in radio technology over the past several decades have made it possible for human civilization to communicate over interstellar distances. Using our own capability as a guide, we can set meaningful lower limits on the characteristics of radio transmissions from ETI sources, for which we might search.

Why radio signals?

Visible light, the most familiar form of electromagnetic (EM) radiation, represents only a minute fraction of the broad range of wavelengths that constitute the electromagnetic spectrum. Other familiar forms of electromagnetic energy are gamma rays, X-rays, ultraviolet light, infrared, and radio waves (including microwaves). All of these types of radiations travel at the speed of light. They all result from abrupt movements of electrons, occurring at greater or lesser energies, which generate shorter or longer wavelengths, respectively (Figure 1). In general, the longer the wavelength of EM radiation, the easier it is for it to cross interstellar distances without being absorbed or deflected by the gas and dust that permeate the plane of our Galaxy. Thus radio waves from the far side of the Galaxy are virtually unimpeded in some directions where visible light is totally obscured.

A formidable problem for beings sending a signal to recipients whose technology is unknown is selection of wavelengths that the recipients will somehow notice. If the intended recipients have eyes, they might well see a powerful visible-light laser beam aimed in their direction. If they have radio telescopes, there are certain wavelengths that astronomers everywhere, ETI or terrestrial, are likely to be monitoring. Two such wavelengths include microwaves with wavelengths of 21 and 18 cm, emitted respectively by spontaneous changes in interstellar hydrogen atoms (H) and hydroxyl ions (OH). Although many other atomic and molecular species have been observed at hundreds of specific wavelengths, the scientific significance of H and OH is compelling enough that radio telescopes everywhere are likely to be tuned to them. An artificial signal at (or near) one of these wavelengths sent in the direction of ETI’s who are

practicing radio astronomy might well be noticed, even though they are not deliberately seeking artificial transmissions.

In addition to the fact that EM radiations cross interstellar space at the fastest achievable speed (the speed of light), they have another feature that makes them attractive to SETI researchers. That is, a civilization may release them unintentionally. Our planet has been inadvertently broadcasting radio waves, TV carrier waves, radar transmissions and microwaves into space for more than 50 years. As a result, the Earth is at the center of an expanding sphere of EM transmissions that is now 100 light-years in diameter and expanding outward at the speed of light. This invisible sphere now encloses about 10,000 stars, our nearest neighbors in space. If there are inhabited planets orbiting any of these stars, their inhabitants might have detected the presence of human technology on Earth. Should they aim sensitive radio receivers in our direction, they would find dozens of radio and TV EM carrier wavelengths and sporadic radar signals that are at least a million times “brighter” than the radiation coming from our star at comparable wavelengths.

The Microwave Search

The preceding considerations have prompted NASA researchers to scan a particular set of EM wavelengths (microwaves) as the most promising way of beginning a search for evidence of ETI civilizations. Microwaves, like all EM radiations, cross interstellar space at light speed. They can carry information, and they can reveal a civilization’s presence even if that civilization has not chosen to send deliberate transmissions. In searching for them, we can scan a much larger fraction of the galaxy than would be the case if we chose to search for physical artifacts.



Frank Drake, whose 1960 experiment, Project Ozma, initiated the modern search for extra-terrestrial intelligence.

This range of frequencies between 1,000 and 10,000 MHz is called the “microwave window.” It happens that natural background radiations that could mask a genuine ETI signal are weakest in this range of wavelengths, making the universe more “transparent” to the passage of signals at these wavelengths than at any others. Special attention is given to the microwave region between the 21 and 18 cm emissions from H and OH, sometimes referred to as the “water hole” (after H-OH, H₂O).

Microwaves were first suggested as promising EM wavelengths for a SETI search some 32 years ago by Cocconi and Morrison (1959). Drake, working independently, conducted the first search for ETI microwave transmissions

in 1960, in an effort now known as Project Ozma (Drake 1960).

Although the preceding discussion has focussed upon *wavelengths* of EM transmissions, in practice SETI researchers usually describe the *frequencies* of the transmissions. The frequency, which is the number of waves passing each second, is related to the wavelength and is simply an alternative feature that may be used to describe EM radiation. The frequency corresponding to one wavelength per second is one Hertz (Hz). For microwaves, typical units are a million and a billion wavelengths per second, one megahertz (MHz) and one gigahertz (GHz), respectively. The "special" frequencies of the 21 and 18 cm microwaves are 1420 MHz (hydrogen) and 1662 MHz (hydroxyl), respectively. In keeping with the custom of the researchers, the rest of this text adopts frequencies as the preferred descriptor of EM radiation.

The frequencies that SETI researchers will scan in the forthcoming NASA HRMS survey are those between 1,000 and 10,000 MHz. This range includes the frequencies used by terrestrial radar and communication transmitters, and also anticipates frequencies that ETI's might reasonably be expected to use if they were seeking to maintain or initiate communication over interstellar distances. Thus, the frequency range selected by NASA has many advantages that include the possibility of eavesdropping, of detecting non-communicative technology, and of detecting signals deliberately beamed in our general direction.

How and Where to "Look" for Microwave Transmissions

Looking For The First Signal

The characteristics of a potential incoming ETI signal are not determinable in advance of detection. However, selecting a plausible band of frequencies (and other signal characteristics) that might be used for interstellar communication helps to reduce the enormity of the initial search and keeps the cost of the receiving hardware affordable. The list of unknown signal parameters includes the transmission frequency, the time of arrival, the strength (or, more likely, the weakness) of the signal, the direction from which it arrives, and features like its polarization or possible pulse characteristics. Exactly how a deliberate signal might encode information is also unknown. A search for signals is sure to be confused by the barrage of EM radiations generated by natural processes in the cosmos and by humanity's own signal transmissions. Local transmissions from the earth and from orbiting satellites are likely to be thousands or millions of times stronger than any natural or artificial interstellar transmissions that we might detect.

To achieve the high sensitivity needed for picking up very faint signals, radio telescopes are designed to have very small "fields of view." For example, a typical radio telescope working at microwave frequencies only receives energy from a small spot on the sky a few minutes of arc in diameter. This "patch" of observed sky is about ten times smaller than the apparent diameter of the full moon (whose diameter is 30 minutes of arc). With such a tiny field of view (comparable to the patch of sky one would see looking through a soda straw), searching for signals from thousands of "nearby" stars or from the entire sky

must necessarily take a long time!

All of these factors strongly influence the designs of the receivers and signal processors needed to collect and recognize a transmission of ETI origin. The program of observation must be carefully planned to target each star and to survey the entire sky as efficiently as possible to avoid wasting "telescope time," which is a valuable commodity to radio astronomers.

The uncertainties dictate the following:

- a) that the antennas be very large to collect as much of the weak signal as possible (or that a linked array of several smaller antennas be used);
- b) that the antennas spend as much time as is practical "looking" at each direction or sector of the sky;

- c) that the antennas be programmed and calibrated to scan small areas of the sky and produce mosaic maps of the entire sky.
- d) that the receivers analyze a wide range of frequencies;
- e) that the special purpose signal processors and computers discriminate between a verifiable signal of ETI origin and various naturally-occurring radio noises from the universe;
- f) that the search procedures distinguish between a verifiable signal of ETI origin and various transmissions of human origin.

The reasons for most of these requirements are explicit in the preceding discussion. The last points, however, bear further explanation.

The universe is awash with background radio "noise" that can easily obscure a faint signal of ETI origin. However, virtually all sources

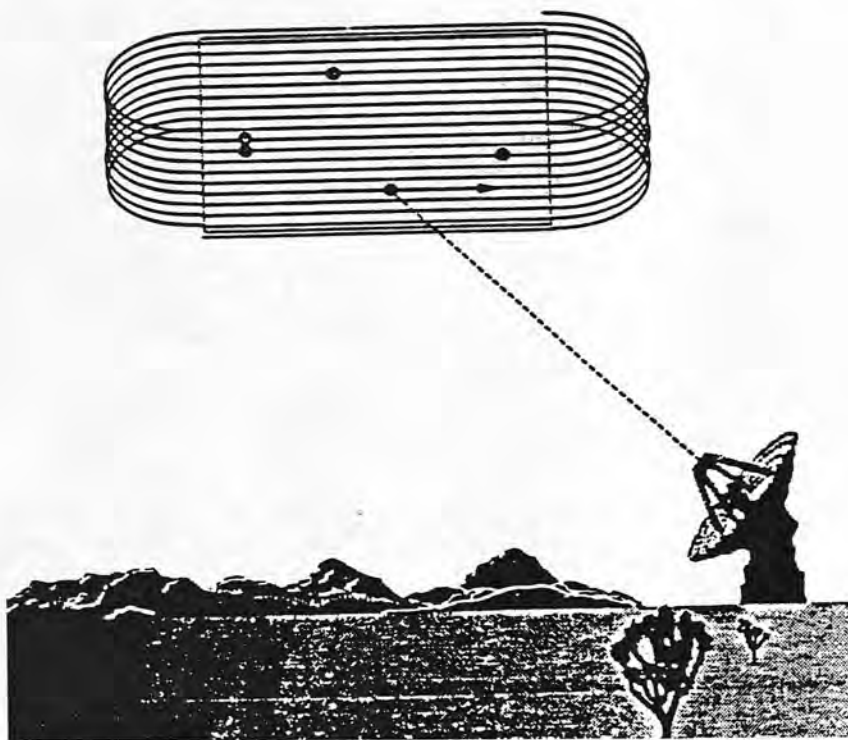


Figure 2. The pattern of observations envisaged for the Sky Survey.

of natural “noise” have a distinguishing feature; they are “broad band.” That is, stars, clouds of gas and dust, pulsars, black holes, supernovas and a host of other objects simultaneously emit EM radiation over a broad range of frequencies. By contrast, transmitters made by human engineers are typically very “narrow band” devices designed to concentrate large amounts of power in a very compact frequency range. Looking for narrow-band signals is analogous to tuning a radio across the band during a thunderstorm, trying to hear the faint broadcast of a distant radio station among the crackles of lightning-induced static. The narrowband nature of the transmitted signal, which comes in at only one frequency, helps to distinguish it from the background static, which can be heard everywhere on the radio dial. Similarly, any signal from the cosmos that is highly compressed in frequency would be carrying telltale evidence of its artificial origin. Such a signal could be isolated and identified, despite its weakness, by specially-designed fast-acting computers. These special purpose computers are the central components of the HRMS signal detection systems.

Stray signals from human sources – satellites, airplane radios, microwave transmitters, cellular telephones – are particularly bothersome, since they often generate precisely the kinds of narrowband signals that NASA’s HRMS system seeks. They necessitate a “verification” procedure that has important implications for public response to the project. Verification is a followup action taken to determine whether a newly-detected and promising signal is anything more than a stray transmission from some human activity.

If a candidate signal is observed emanating from some point in the sky, the radio telescope’s control computer will initiate a pointing maneuver that will alternately position the

telescope’s field of view, toward and then away from the sky position of the candidate signal. If the signal detection computer continues to “see” the signal when the telescope is looking in some other direction, then it is evident that the signal is not really coming from the sky, but from some human device located nearby. If the signal disappears - and then reappears when the telescope is pointed back toward where it was first seen - the operator on duty begins additional “on-site” verification tests.

If the candidate signal passes these initial “on-site” tests, the next phase of the verification process commences. A second radio observatory, assigned in advance as the HRMS verification site, would perform similar tests at the sky position and received frequency of the candidate signal. If the signal is detected and passes certain verification tests at the second site, the scientists would conclude that the source of the candidate signal was indeed outside the solar system. If the second observatory fails to detect a signal that is “seen” by the first observatory, this would be regarded as evidence that the “signal” is really of human origin, even though it may have passed the initial screening procedure at the first observatory. If possible, signals that pass all verification tests will be further scrutinized by other independent confirmation sites.

This elaborate verification and confirmation process is important and necessary. However, it also has potential for generating “leaks” of information and “false alarms” that may significantly color the public’s perception of SETI (see Chapter 5).

Studying A Detected Signal

If a signal is detected and verified, it will probably be very easy for scientists everywhere to study. The enormous computer power necessary to sift and search the background noise from the cosmos is not required for followup studies, once the characteristics of a signal and its sky coordinates become known. With information in hand about an arriving signal, astronomers in many countries would probably build simple receivers, tune to

the incoming frequency, and study the signal. The task of observing a known signal is so much easier than finding the signal in the first place that other researchers (and perhaps even dedicated amateur groups) may be able to “tune in” on a strong ETI transmission.

A signal from a planet orbiting a distant star would appear to come from a fixed point in our sky, as unchanging from year to year as the positions of the other stars. That position would appear to rise and set with the stars and

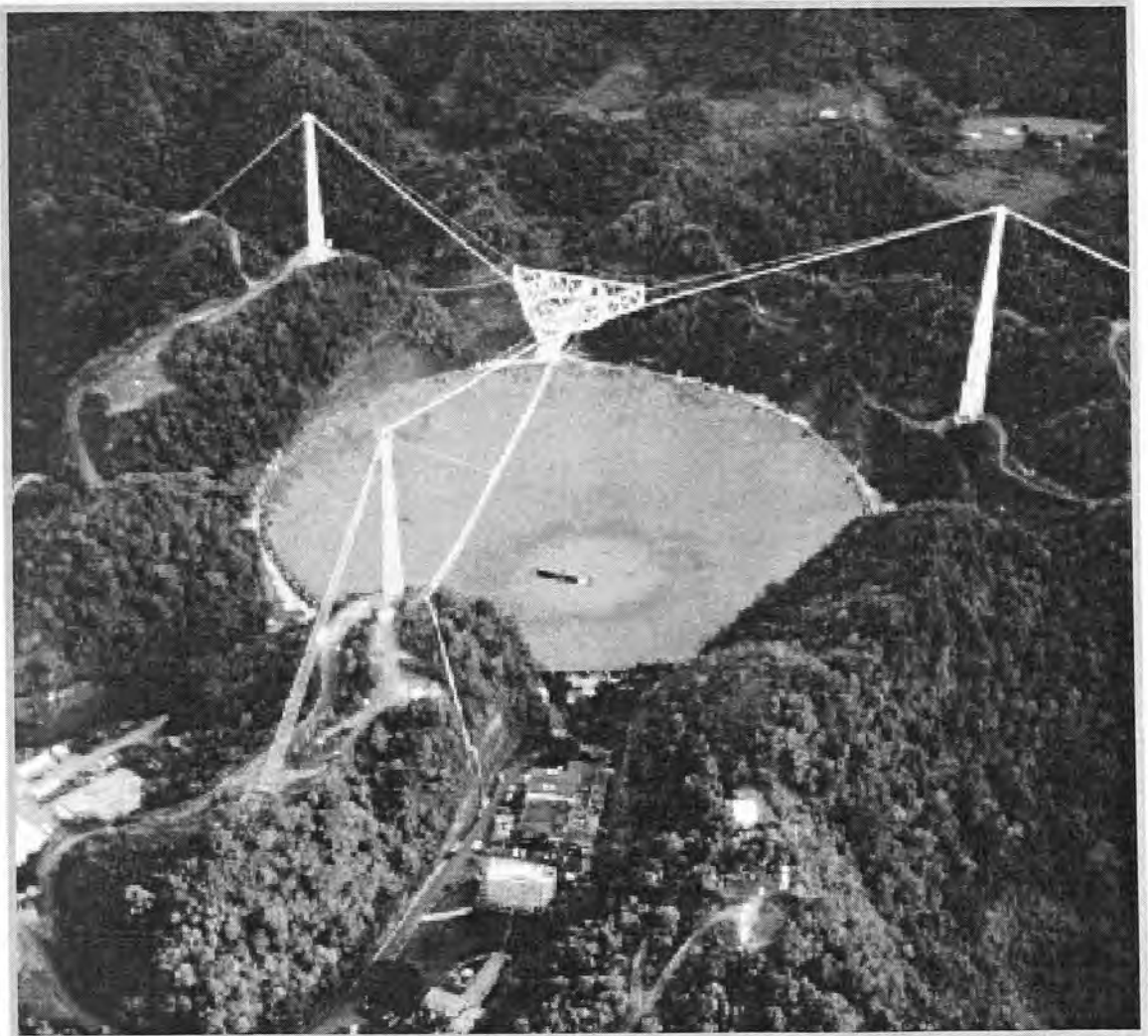


Figure 3. The 1,000 foot diameter Arecibo Radio Telescope, in Puerto Rico. An instrument of unsurpassed sensitivity, this antenna is used for SETI targeted searches.

would be accessible to radio telescopes around the globe by day or night, depending on the season of the year. It is likely, but not assured, that the point from which the signal emanates could be “seen” from most nations on Earth for at least a few hours every day. If the source were immediately overhead to observers on the Earth’s equator, then the signal would be accessible to every country, about 12 hours out of every 24. At the other extreme, if the source were directly over the north (or south) pole, then it would be “visible” only to countries in the northern (or southern) hemisphere (for 24 hours per day). A source located in between these extremes would be visible to most countries for longer or shorter daily intervals. (For example, a source at 45°S would never be visible from sites north of latitude 45°N, but would be visible from all sites south of that latitude, to the south pole.)

This property of the skies and the Earth’s rotation makes it likely that most nations will have some daily access to the signal, with some enjoying much longer daily opportunities for observation than others. The least access would result if the signal source were located in the sky directly over Antarctica.

These considerations have geopolitical implications. If the signal is not repetitive, cooperation and sharing would be required in order for observing nations to pool their shares of the intercepted signal and to reconstruct a complete transmission. However, the likely prolonged opportunities for most nations to observe most of the signal themselves practically assures ease of access to information about it and from it, by any interested nations, groups, or individuals. This is one reason why NASA HRMS researchers consider that any incoming ETI signals truly belong to all of humankind.

The Survey Strategy and Technology

The HRMS project will modify and use existing radio telescopes around the world. To these telescopes it will bring the special-purpose signal processing equipment required to divide incoming radio frequencies into tens of millions of narrow channels and the computers needed to sift through the channels seeking strong narrowband continuous wave or pulsed signals. These systems will usually run unattended, automatically cataloging any candidate signals while the radio telescope is in operation. The need for the computer to recognize a candidate signal hidden amid tens of millions of channels of static, all performed while the telescope is actively engaged with the search, requires an extremely fast signal processing system. The requirements are made more demanding by the need for signal analysis and automated screening of Earth-based sources of radio interference. The computational speed of the HRMS systems exceeds the speeds of state-of-the-art supercomputers. The HRMS systems achieve these prodigious speeds because they are special-purpose devices that do not require the complexity and versatility of a general purpose computer.

Two complementary strategies, a Targeted Search and a Sky Survey, are being pursued by the HRMS researchers. These strategies will result in a systematic and comprehensive search of the sky that surpasses by several orders of magnitude the sum of all previous SETI efforts. The Targeted Search, managed by NASA’s Ames Research Center, uses some of the world’s largest radio telescopes to search for signals from the vicinity of relatively nearby “target” stars that are similar to our Sun’s age, size and brightness. The Sky Survey team, based at the Jet Propulsion

Laboratory in Pasadena, is working to survey the entire sky by using the smaller (34-meter diameter) antennas that are part of NASA's global Deep Space Network (DSN). Observations by both approaches began in October, 1992, and are scheduled to continue for a decade. These years of observation are needed to survey the vastness of space and sift through hundreds of millions of discrete frequencies to seek out candidate signals.

The Targeted Search will observe nearby stars that may, by virtue of their similarity to the Sun, be likely to host planets like our Earth, with evolved life. Because the telescopes need not survey the whole sky (but only the selected target stars), it is possible for this search to devote more time to each target, with more detailed analysis of the wavelengths coming from each, than is possible in the Sky Survey. The Sky Survey, with more area to search, will spend less time on each sector of sky but will ultimately scan the entire celestial sphere, including all the spaces between the target stars examined by the Targeted Search. Thus, the Targeted Search is well suited for picking up weak pulsed or continuous signals from civilizations at selected nearby locations in space, whereas the Sky Survey is well suited for detecting more powerful continuous transmissions, wherever their source (Figure 2).

Together these two search strategies will accomplish a search that is millions of times more comprehensive than anything that has been done, or was even possible, over the past 30 years.

The Targeted Search

The Targeted Search concentrates on the 1,000 to 3,000 MHz portion of the

microwave window, where the best sensitivity can be achieved. This band of frequencies matches the microwave window at the lower end. The 3,000 MHz limit was also selected to match the performance limits of the world's largest radio telescopes, most notably those at Arecibo and Nançay. The performance of these (and other) large radio telescopes decreases at higher frequencies because of



John Billingham, Co-chair of the Working Group and Head of the NASA SETI program at Ames Research Center, Moffett Field, California.

irregularities in their surfaces. Higher frequencies may be observed as opportunities arise, however, at other observatories.

The current list of target stars contains 773 stars chosen from the *Royal Greenwich Observatory Catalogue of Stars Within 25*

Parsecs. These were selected for their similarity to the Sun (Wooley and others, 1970). The list is now being optimized to remove multiple star systems (which are unlikely to have stable planetary orbits), stars that are deemed to be too young for the billions of years of evolution that are thought to be necessary for the rise of higher forms of life on their nearby planets, and stars that are very dissimilar to the Sun in size, temperature, or stage of stellar development. Stars that are removed from the list will be replaced by more promising candidates. When the task is completed, the target list will contain about 1000 solar-type stars within 100 light-years of Earth. These nearby stars will be approximately uniformly distributed in the sky and will require that telescopes in the northern and southern hemispheres be used to conduct the observations.

Each target star will be observed for at least 300 seconds at every frequency between 1GHz and 3 GHz. Stars closer than about 20 light-years will be observed for at least 1000 seconds at every frequency (as will all stars observed from a facility to be dedicated to the project at Green Bank). This plan increases sensitivity to signals from any civilizations near enough to have detected (and responded to) our own television or radar transmissions, and it also allows for the detection of pulsed signals with long pulse repetition periods.

Antenna collecting area (= "size") is the most critical factor in determining the sensitivity that can be achieved in a given observing time. Therefore the Targeted Search will utilize the largest available radio telescopes. The Targeted Search expects to use about 5% of the available observing time on the world's largest antenna, the 1000-foot Arecibo radio telescope in Puerto Rico (Figure 3). Around 1995, the Targeted Search will assume operation of the National Radio Astronomy

Observatory's 140-foot telescope at Green Bank, West Virginia as a dedicated facility for SETI observations. Observing time on the 64-m diameter antenna at Parkes, Australia (which is operated by the Commonwealth Scientific and Industrial Research Organization – CSIRO) has been negotiated with the Australian Space Agency. In addition, it is possible that a 91-m telescope at NanÁay, France and a 70-m antenna at NASA's DSN Complex at Tidbinbilla, Australia, will be used to complete the observations.

In all, about 50,000 hours of observing time are required to complete the Targeted Search.

The Sky Survey

The objective of the Sky Survey is to search the entire sky over the frequency range from 1,000 to 10,000 MHz. The choice of NASA's 34-m antennas is a compromise between sensitivity and the time required to survey the whole sky. (Smaller antennas with larger fields of view can scan the sky more rapidly than large antennas at a given frequency, but are also less sensitive.) The control computers are programmed to scan the sky in predetermined patterns, and are calculated to finish the survey in about six and a half years, once the full-up system is operational in the mid-1990's. The Sky Survey has requested up to 16 hours per day of observing time on one of the 34-m radio telescopes in NASA's Deep Space Network, with support from other radio telescopes of comparable size and sensitivity. A prime example is the 39-m radio telescope at the Owens Valley Radio Observatory operated by the California Institute of Technology. The northern sky will be surveyed first using the new 34-m radio telescope at the DSN 13 Research Station at the Goldstone Deep Space

Communications Complex (Figure 4). Later, Sky Survey equipment will move to Tidbinbilla, Australia, to complete the surveys of the southern sky.

Small areas of the sky, called sky frames, are individually surveyed and then form mosaic maps of the entire sky. Each sky frame is examined 31 times as the receiving systems are stepped through a series of 31 adjacent frequency bands, each 320 MHz wide. Each sky frame takes about one hour to complete. Approximately 25,000 scans are required to cover all the frequencies and all directions in the survey.

Because the Sky Survey scans a broader range of frequencies with smaller telescopes than is the case with the Targeted Search, its data processing requirements are similar but not identical. In particular, the back-and-forth sweeps of the telescope as each sky frame is built up allow the opportunity to take “second looks” at promising candidate signals detected on a previous pass. The signal processing computers use the additional information to cull candidate signals from the background noise and ubiquitous radio frequency interference. Like the Targeted Search, the systems are designed to operate automatically with occasional tasks for the on-site operator or the on-call scientist.

The Future of the Search

All foreseeable outcomes of the HRMS survey will be of great intellectual and emotional interest to humanity. The research will probably produce important scientific and technological spinoffs, as well. This final sec-

tion describes some of those possibilities as they relate to the Workshops’ task – the assessment of human responses to the search and its possible outcomes.



Dave Milne, Science Education Coordinator, SETI Institute, and Executive Secretary of the Workshops.

A Dialog With Extraterrestrial Beings?

The previous text has referred to the detection of signals of ETI origin and has hinted that a variety of different signal types might be anticipated. These include continuous carrier waves and narrowband pulses. Like the carrier frequencies that encode our television broadcasts, the artificial origin of such radiation would be easy to recognize, but any information in it – music, pictures, news – would be very difficult to recover because it would be weaker than the carrier and would need to be demodulated (decoded) in order to

be interpreted. Such carrier waves (and emissions from radars and other devices that had no information encoded in them at the source) would be similar to a "dial tone," signifying that some technology is sending the signal. Like a genuine dial tone, such a signal would have no encoded information content. At the other extreme, transmission of information across space may be intended by the senders. If transmitting ETI species are communicating among themselves, using a code or language that both senders and recipients already understand, then it might be very difficult (or impossible) for us to decode intercepted messages that were not intended for us. However, if the senders' signals are directed at recipients who are not expected to know their language, an encoding method might be selected that is easily recognized and "solved" by any recipients, even in the absence of a language common to receivers and senders. The first signal might well contain simple instructions for improving the recipients' ability to intercept and translate any information that follows. In crafting a message intended for beings who are new to the communication network, we can envision ways in which the transmitting civilization might use "anti-cryptography." That is, the ETI senders could make a message directed at strangers that is as easy as possible to decode. In such a case, the first part of the signal would contain "easy" and "obvious" instructions on how to read the rest.

We know of two authors who have tackled the task of developing "interstellar languages." They speak from a human standpoint, and envision an anti-cryptographic language for use in transmissions from Earth to ETI's. In fact, humans would be the recipients of a message in the NASA SETI endeavor; HRMS will not transmit any signals or reply if a signal is detected. However, the exercise helps us

think about the decoding process. How would we do it?

In 1960, Freudenthal developed an interstellar language in great detail. He started by making no assumptions about the scientific and technological capability of the recipient civilization, and built up a lengthy and laborious tutorial that begins with simple mathematical concepts (Freudenthal, 1960). DeVito (1990) took a somewhat different approach. He made the assumption that the other civilization would be at least as advanced as ours in science and technology, and proceeded through a logical exposition of the basics of science, with emphasis on explaining those quantities that we on Earth have defined in an arbitrary manner. Numbers come first, followed by many other scientific concepts that (he conjectures) would be readily recognized by scientifically-trained ETI observers. Descriptions of the physical and chemical structure of our planet and its biology follow. When it comes to human physiology, behavior, history, and sociology, explanation becomes much more difficult. DeVito proposes (as have others) to combine the language with pictures, which are literally "worth a thousand words" to beings with visual perception. Pictures are particularly helpful when describing complex engineering projects. These studies suggest that fairly complete "self-explanatory" interstellar languages can be devised. If we detect a signal, and it incorporates a message, we might hope that it would contain a comparable interstellar language tutorial, if the transmission is intended for us.

If a self-explanatory message arrives from ETI's, laden with information, the consequences for humanity would surely be profound. Exactly what those consequences might be would depend, of course, on the content of the message. Anticipating peoples'



Figure 4. The 34 m DSN telescope at Goldstone, California.

reactions is not easy, but the task is begun in Chapter 3 of this volume.

Near-Term Scientific Benefits

Information obtained with SETI observation-lal systems, and alternative uses of the systems themselves, will probably be of benefit to humanity in areas other than SETI. In the domain of astronomy, it has been suggested that any new instruments that search nearby space in dramatically new ways are likely to discover unexpected major new astronomical phenomena. (This has been generally true over the last fifty years.) The NASA HRMS Project is likely to discover new astronomical or astrophysical phenomena other than ETI's (Tarter, 1984). The HRMS equipment can also be used for various radio astronomical projects. The Sky Survey is a good vehicle for continuum source surveys at minimum extra cost. Other possibilities are spectral line surveys and use of the "dwell mode" to enhance the sensitivity for radio astronomy

applications. Further details can be found in the 1983 SETI Science Working Group Report (Drake and others, 1983).

The NASA HRMS digital spectrometer and signal processors also hold promise for use by other disciplines. Any field that can benefit from the high spectral resolution detection of signals amid complex background noise can profit from the new technology. Examples include radar astronomy, aircraft communications and signal analysis, geology and geophysics, mass spectrograph or gas chromatograph design, computer-aided tomography, magnetic resonance imaging, neurophysiology, electroencephalography, and remote sensing systems.

The Future of NASA's SETI Research

No ETI signal has yet been found. The current program, which we may call "HRMS I," may detect one – or it may not. In either case, we will learn something important

about the likely proximity of intelligent neighbors in space.

If a signal is found, there would undoubtedly be intense interest in expanding and intensifying the investigations – an extension that we may call “HRMS II.” It is likely that many nations (and probably even corporations, organizations, and wealthy individuals) would put resources into this effort. These would rapidly advance the sophistication of SETI technology, perhaps even resulting in such grand-scale apparatus as an orchard of antennas on Earth to increase collecting area, or antennas in space or on the far side of the Moon. These efforts would also enlarge our knowledge of the detected ETI civilization.

A verified detection would compel human beings to recognize that we are but one of many civilizations, and are perhaps even “primitive” in comparison with our extraterrestrial neighbors. At the same time we may begin to learn presently-unimaginable new facts about the universe, life, intelligence, and society. Our possible responses to these monumental shifts in our view of life are addressed in the remaining chapters of this report. The questions addressed there, and the likely answers, are profound and require the intensive study that we recommend, beginning now.

There is an alternative outcome to the HRMS I search that is not unlikely. HRMS I may be completed at the end of the decade without detecting a signal. This provides a different type of stimulus for conducting HRMS II. Under these circumstances it would probably be more difficult to obtain support for a continued search. If HRMS II is carried out, and no signal is found, HRMS III may be contemplated sometime after 2010. It is possible (and even likely) that these searches may take a very long time. We should not be timid

about our quest to learn whether we are alone in the universe; we should resolve to press on until we discover an ETI transmission or learn enough to conclude that we are alone.

In the latter event, humanity will face other philosophical questions, perhaps as profound as knowledge of the existence of ETI's. Do civilizations always bring about their own extinction? Is the vast silence of the cosmos stark evidence of that fact, a warning to us? Or is intelligence so rare that civilizations seldom start up? To begin to answer these questions, – we must search the skies.

We conclude with a statement known as the “spade analogy,” by Philip Morrison, in which he compares SETI research technology with the tools by which archaeologists investigate humanity's past.

“...I think the SETI enterprise can best be understood as an exercise in the archaeology of the future. We are well aware of the archeology of the past: we find a site, a tumulus, or a ruin, and we take a spade and we dig into the ground, and if we are lucky, we discover Ur of the Chaldees or something marvelous! We never thought that we could examine the same thing in reverse time. But in fact, in a way, we can. [By detecting another civilization] it is of course their past we're investigating... but our future. Maybe the spade will turn up a good site one day. We hope it will. It's just a question of being patient.

When you've got the spade and you know the future is there, it seems very wrong not to dig...”

(Philip Morrison, Institute Professor of Physics Emeritus, Massachusetts Institute of Technology)